Lecture 3

Probability & Maximum Likelihood

September 22, 2025

Previous Lecture

- Fitting a model to data e.g. $\hat{y} = \theta_0 + \theta_1 x$
  - Describes relationship we see in the data (between x and y).
  - We learned the OLS method, where $\left(y - f_{\hat{\theta}}(x)\right)^2$ gets optimized.
  - Parameters $\hat{\theta}$ found by minimizing a least squares loss function.

ML Recap

ML Recap

x	*y*
5	1
9	5
6	2
12	8
13	9
7	3
9	5

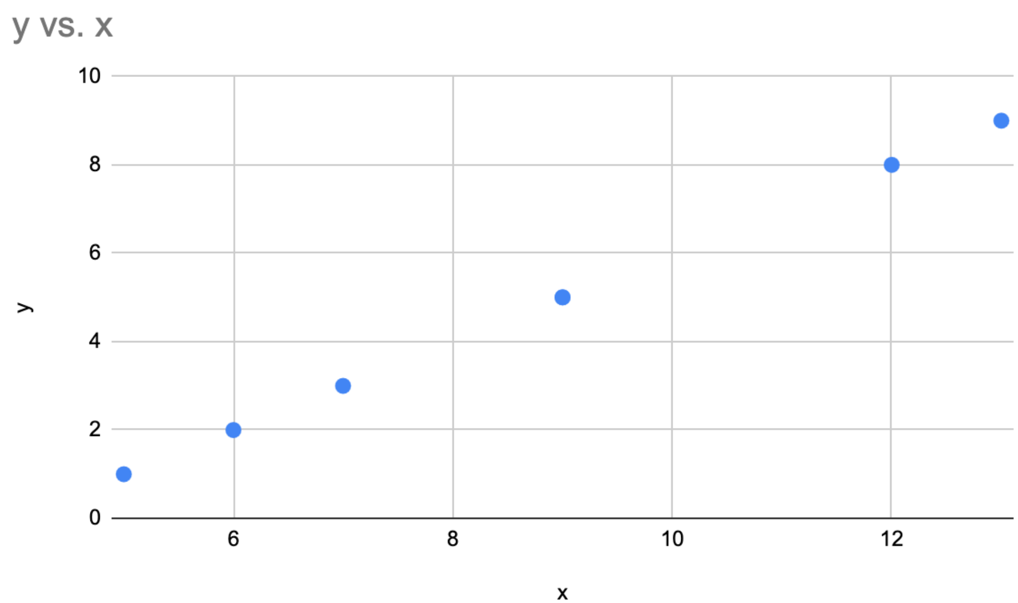
-We have some data, and that'd like to make predictions with.
-We will want to use features from the data x to predict y.

-Ex. Temperature (x) predicting Power Consumption (y).
-Ex. Number of Rooms in a house (x) predicting the Price of a house (y).

ML Recap

-After plotting the data, we see that the relationship between x and y looks linear, so we decide that we want to model using linear regression.

x	*y*
5	1
9	5
6	2
12	8
13	9
7	3
9	5



ML Recap

x	*y*
5	1
9	5
6	2
12	8
13	9
7	3
9	5

$$\hat{y} = \theta_0 + \theta_1 x$$

-We can start by choosing random values for our parameters and a value from our dataset, and we can start making predictions.

ML Recap

x	*y*
5	1
9	5
6	2
12	8
13	9
7	3
9	5

-We randomly choose $\theta_0 = 3$ and $\theta_1 = 4$.

$$\hat{y} = 3 + 4 * \boxed{5}$$

ML Recap

-We can compare our prediction $\hat{y}$ with the actual y values in our dataset to know if our model parameters are any good (they aren't yet).

x	*y*
5	1
9	5
6	2
12	8
13	9
7	3
9	5

$$\hat{y} = 3 + 4 * 5 = \boxed{23}$$

ML Recap

x	y
5	1
9	5
6	2
12	8
13	9
7	3
9	5

-Maybe we randomly choose $\theta_0 = 2$ and $\theta_1 = 1$.

$$\hat{y} = 2 + 1 * \boxed{5}$$

ML Recap

x	*y*
5	$\boxed{1}$
9	5
6	2
12	8
13	9
7	3
9	5

-We randomly chose $\theta_0 = 2$ and $\theta_1 = 1$.

$$\hat{y} = 2 + 1 * 5 = \boxed{7}$$

$$\hat{y} = 3 + 4 * 5 = \boxed{23}$$

-Our last attempt with $\theta_0 = 3$ and $\theta_1 = 4$ was clearly worse.

ML Recap

x	y
5	1
9	5
6	2
12	8
13	9
7	3
9	5

$$\hat{y} = 2 + 1 * 5 = \boxed{7}$$

$$\hat{y} = 3 + 4 * 5 = \boxed{23}$$

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad \text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

-More formally, we can check which parameters performed better using a cost function like RSS or MSE. These functions aggregate the loss (error for a single data point) across an entire dataset.

ML Recap

x	*y*
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$$\hat{y} = \theta_0 + \theta_1 x$$

-What we want is to find the values of these parameters θ_0 and θ_1 that minimize the model's overall error.

-To do so we need either:

a) **Normal equation** = exact, closed-form, but expensive for high dimensions.

b) **Gradient descent** = approximate, iterative, but scales much better.

ML Recap

$$\hat{y} = \theta_0 + \theta_1 x \qquad RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

x	*y*
5	1
9	5
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12	8
13	9
7	3
9	5

-Let's move forward with gradient descent (GD) and RSS as our cost function for now.

ML Recap

$$\hat{y} = \theta_0 + \theta_1 x \qquad RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

x	y
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6	2
12	8
13	9
7	3
9	5

$$\theta_{iteration\ t+1} \leftarrow \theta_{iteration\ t} - \alpha \nabla g(\theta)\,|_{\theta_{iteration\ t}}$$

-GD iteratively improves model parameters by moving them in the direction that reduces the cost function (RSS). At each step, we compute the gradient of the cost function with respect to the parameters, and update using the learning rate α.

ML Recap

x	y
5	1
9	5
6	2
12	8
13	9
7	3
9	5

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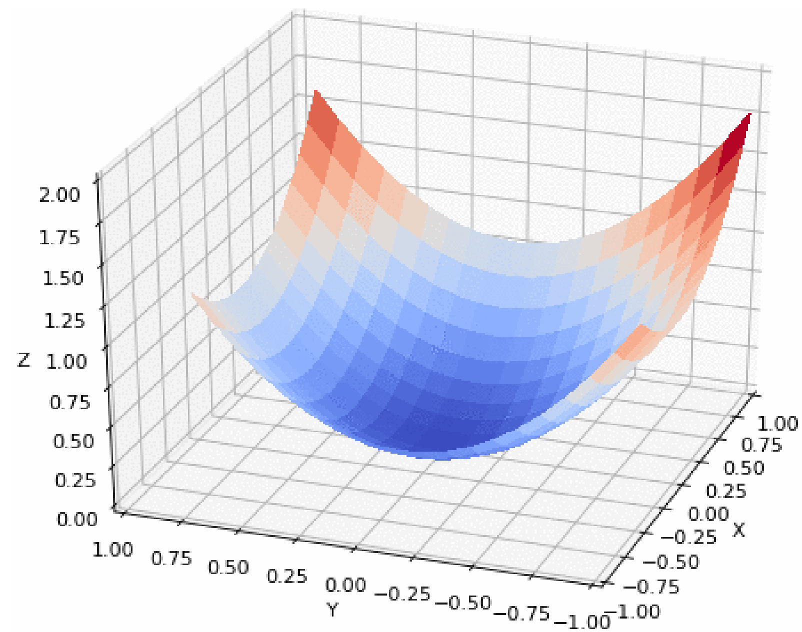

-Repeating this process many times eventually gets us to a point where we can stop after our error is converges to near 0.

Today's Lecture

- (Re-)introduce the language of probability to address a key question:
  - How can we choose our loss function?

- Foundation will also let us examine and evaluate the ability of our model to generalize in future lessons.

3.1: Probabilities and Events

Math in Machine Learning

- Machine learning is programming by **optimization**.
- We need math to **understand that optimization**.
  - **linear algebra**, to understand the objects being optimized (data and models are arrays).
  - **calculus**, to understand how we optimize.
  - **probability and statistics**, to reduce surprise/uncertainty.

Surprises Happen in ML

- Surprise is "inverse probability" and it is foundation of information theory.
- If something is a little **less probable**, it happening is a little more surprising.
- If something is **certain**, it happening is not at all surprising.
- If something is **impossible**, it happening is more surprising than anything else.
- The best model is **least surprised** by the outcome.
- Probability theory can be applied to any problem involving uncertainty.
- In ML, uncertainty comes in many forms:
  - what is the best prediction (or decision) given some data?
  - what is the best model given some data?
- We use techniques such as maximum likelihood for model estimation.

Core of Machine Learning

1. Model parameters is an array $\boldsymbol{\theta} = [\theta_0, \theta_1, \ldots]^T$

2. We update the parameters over time by calculating the gradient of loss (cost) function:

$$\boldsymbol{\theta}^{(\tau+1)} \leftarrow \boldsymbol{\theta}^{\tau} - \alpha \cdot \nabla g(\boldsymbol{\theta}^{\tau})$$

3. By changing parameters, probability of data is changed and we want to minimize the surprise or maximize the probability (find the **most likely** θ that makes the observed data the most probable).

$$Minimize\ Loss(\boldsymbol{\theta}) = Minimize\ Loss\big(y, f_{\hat{\theta}}(x)\big)$$

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When we adjust parameters with gradient descent, we are really trying to **maximize the probability of the observed data** (make the model fit the data better), which is the same as **minimizing the total surprise**.

$$Minimize\ Loss(\boldsymbol{\theta}) = Minimize\ Loss\big(y, f_{\hat{\theta}}(x)\big)$$

Probabilistic Models & Probabilities to Data

- Suppose your model has parameters θ. For each input–output pair (x,y), the model says:

$$p_\theta(y \mid x)$$

- This is the probability of seeing label y given input x, according to the model with parameters θ.
- But isn't linear regression deterministic? Given some parameters how can we have random outcomes?
  - Because data isn't perfect, some values of x may have multiple y's, resulting in the probably of some value y given x.

Probabilistic Models & Probabilities to Data

- Suppose your model has parameters θ. For each input–output pair (x,y), the model says:

$$p_\theta(y \mid x)$$

x	*y*
5	2
5	2
6	2
12	8
5	1
7	3
5	2

Probabilistic Models & Probabilities to Data

- Suppose your model has parameters θ. For each input–output pair (x,y), the model says:

$$p_\theta(y = 2 \mid x = 5) = ?$$

x	*y*
5	2
5	2
6	2
12	8
5	1
7	3
5	2

Probabilistic Models & Probabilities to Data

- Suppose your model has parameters θ. For each input–output pair (x,y), the model says:

$$p_\theta(y = 2 \mid x = 5) = 0.75$$

x	*y*
5	2
5	2
6	2
12	8
5	1
7	3
5	2

Core of Machine Learning

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$$\boldsymbol{\theta}^{(\tau+1)} \leftarrow \boldsymbol{\theta}^{\tau} - \alpha \cdot \nabla g(\boldsymbol{\theta}^{\tau})$$

Gradient descent adjusts parameters to maximize the likelihood of the observed data: for probabilistic models, this learns a distribution; for linear regression, it predicts the mean of y given x.

$$Minimize\ Loss(\boldsymbol{\theta}) = Minimize\ Loss\big(y, f_{\hat{\theta}}(x)\big)$$

3.2: Random Variables

Randomness

We often assume that we can treat items as if they were distributed "*randomly*." The word "random" is used in lots of ways:

- Result of a coin flip is "random"
- Passengers were screened "at random"

A Random Variable is a quantity produced by a random process

Probabilities for Random Variables:

Random variable X, realization x

A **realization** is basically a **specific observed value of a random variable**.

- What is the probability we see x ? $\mathrm{P}(\mathrm{X}=x)$
- What is the probability that we see a positive realization ? $\mathrm{P}(\mathrm{X}>0)$

Sample Spaces and Event Probabilities

- Sample space is the set of all possible outcomes we might observe. Depends on the context
  - Discrete: a finite set of states:
    - Coin flips: $\mathcal{S} = \{h, t\}$
  - Continuous: a range of numerical values
    - Person's height in cm: $\mathcal{S} = \mathbb{R}^{\geq 0}$

- An event is a **subset** of the sample space.
- Any event can have a probability between 0 and 1 (inclusive).
    - $p(\{h\}) = 0.5$
    - $p(height[170, 180]) = 0.10$

- A **random variable** is a mapping from event space to a number (or vector) such as the number of heads that appear among our 10 tosses.
- **Distribution:** a function that shows possible values for a variable and how often they occur.

Sample Spaces and Event Probabilities

The **sample space** of rolling a dice is: {1, 2, 3, 4, 5, 6}

A **realization** would be when you roll the dice and get a 4. It's a single outcome that actually occurs when you perform the task.

An **event** a set of outcomes, can include one or many events. Ex. Rolling a dice three times: {5,1,6}

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Sample Spaces and Event Probabilities

A **random variable** is a mapping from event space to a number (or vector) such as the number of heads that appear among our 10 tosses.

An **event** a set of outcomes, can include one or many events. Ex. Flipping a coin 10 times is an event: {H,T,T,H,T,H,T,H,H,H} and the random variable would map the number of heads that appear in that event to 6.

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Sample Spaces and Event Probabilities

Distribution: a function that shows possible values for a variable and how often they occur.

A **distribution** describes a **random variable** by specifying:

1. The **possible values** it can take.
2. The **probabilities** or **frequencies** associated with each value.

Ex. Random variable: X = number from rolling a six-sided dice.

1. Possible values are x={1, 2, 3, 4, 5, 6}
2. Probabilities are {1/6, 1/6, 1/6, 1/6, 1/6, 1/6}

Ex. Random variable X = number of heads in two coin flips.

1. Possible values are x={0, 1, 2}
2. Probabilities are {¼, ½, ¼}

Event Space: {T, T}, {T, H}, {H, T}, {H, H}

Sample Spaces and Event Probabilities

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Event Space: {T, T}, {T, H}, {H, T}, {H, H}

3.3: Discrete Random Variables

Example 1

Discrete Random Variables

- Discrete random variables take values from a countable set
  - Rolling a die
  - Six sides 1, 2, 3, 4, 5, 6

Outcome	Probability
1 ; 2 ; 3 ; 4 ; 5 ; 6	$\frac{1}{6} = 0.17$
7	0

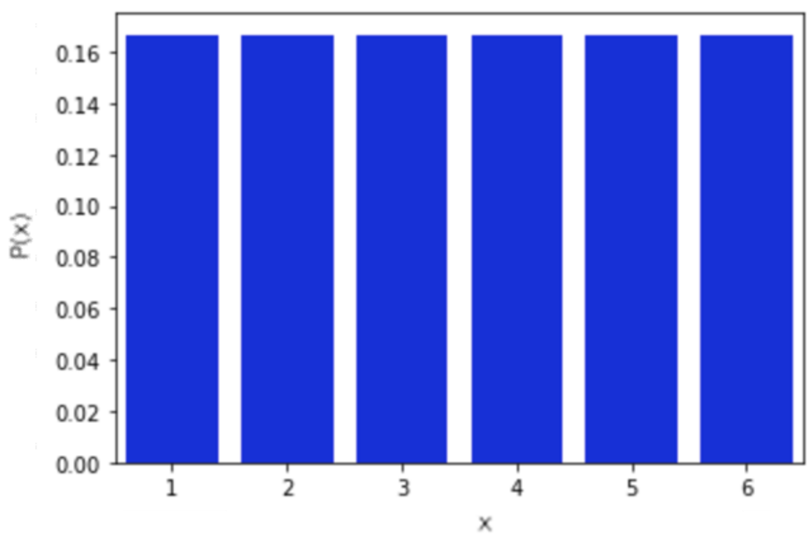
  - Rolling two dice
  - 36 possibilities

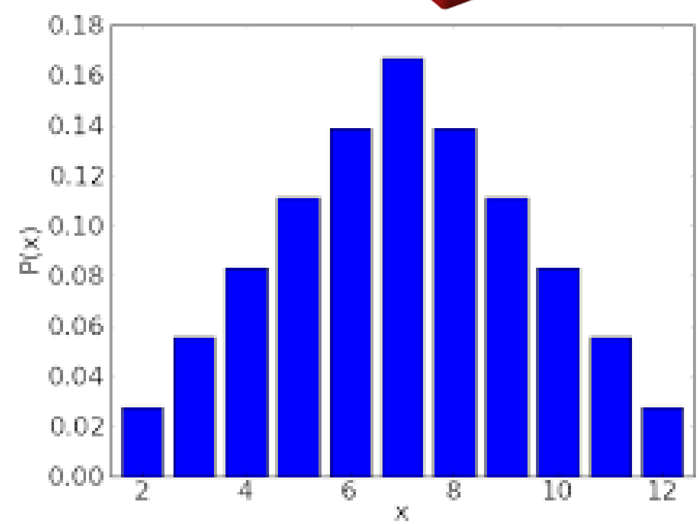
(1,1)	(2,1)	(3,1)	(4,1)	(5,1)	(6,1)
(1,2)	(2,2)	(3,2)	(4,2)	(5,2)	(6,2)
(1,3)	(2,3)	(3,3)	(4,3)	(5,3)	(6,3)
(1,4)	(2,4)	(3,4)	(4,4)	(5,4)	(6,4)
(1,5)	(2,5)	(3,5)	(4,5)	(5,5)	(6,5)
(1,6)	(2,6)	(3,6)	(4,6)	(5,6)	(6,6)

Outcome	Probability
1 ; 2 ; 3 ; 4 ; 5 ; 6; 7; 8; 9; 10; 11; 12	0; 0.03; 0.06, …,0.03
13	0

Probability Mass Function (PMF)

- For a discrete X, $p_X(x)$ gives $P(X = x)$
- Requirement: $\sum_{x \in \chi} p_X(x) = 1$, χ is set of possible realizations





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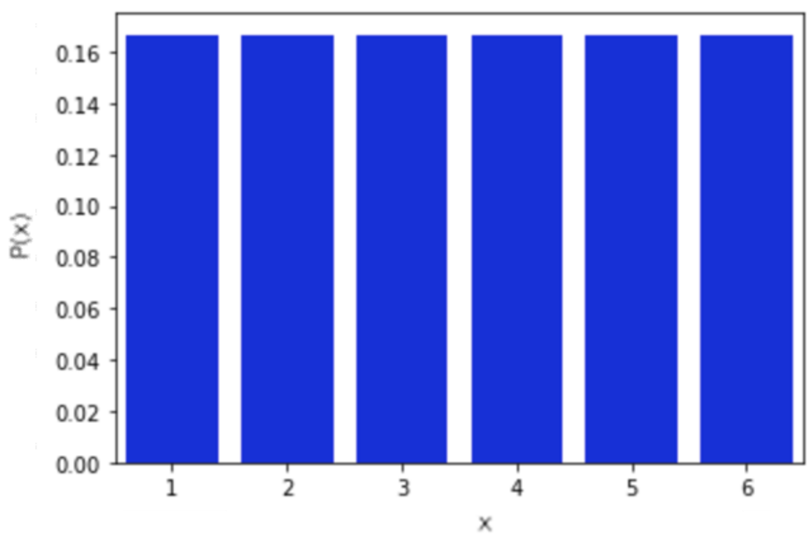
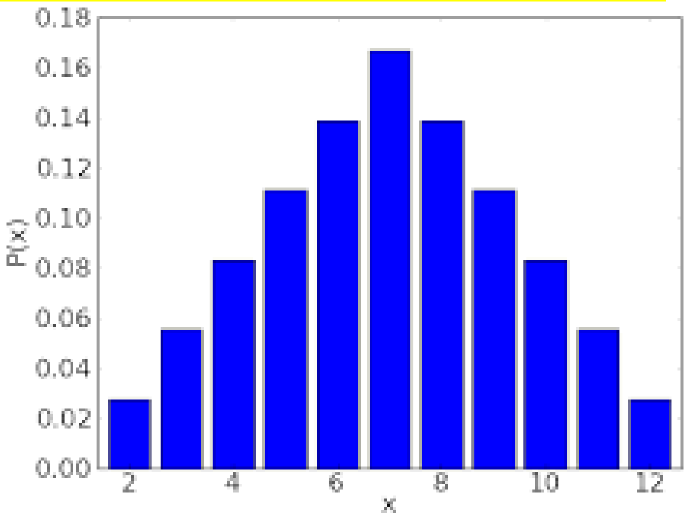
The PMF tells you **how likely each outcome is**. It is a function that assigns a probability to each possible outcome of **discrete random variable**. The probabilities are between 0 and 1, and the sum of all probabilities adds up to 1.





Cumulative Distribution Function (CDF)

The CDF accumulates probabilities from the smallest value up to x.

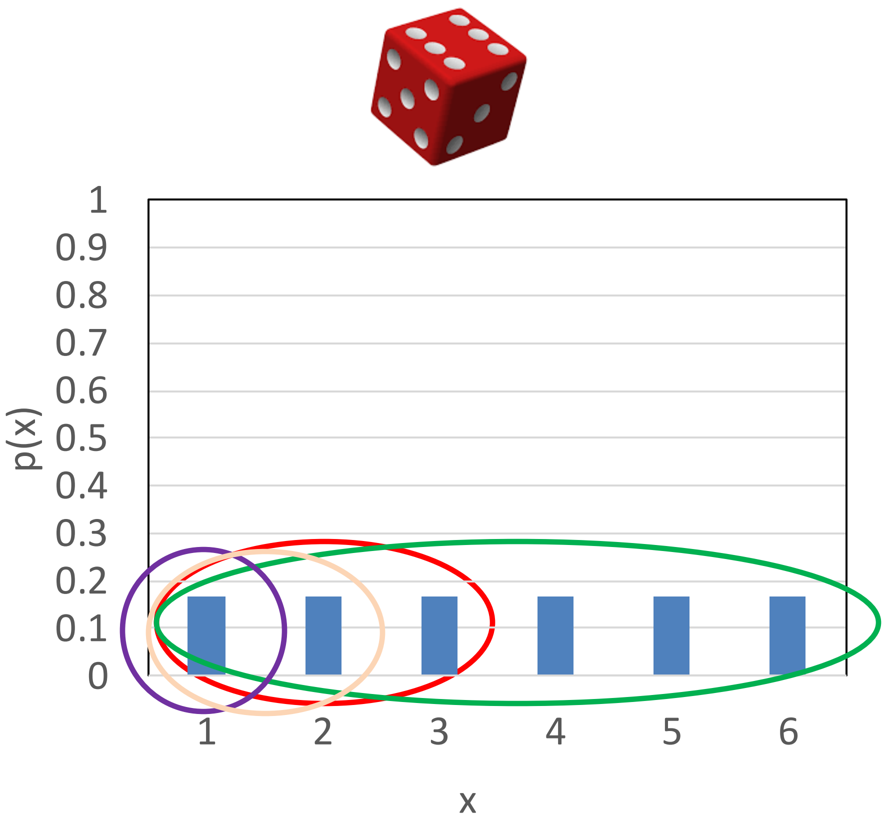
- For a discrete $X, \; P_X(x)$ gives $P(X \leq x)$
- Requirement:
  - $P \; is \; nondecreasing$
  - $sup_{x \in \chi}(P_X(x)) = 1$

Supremum (the least upper bound), as x ranges over all possible values is 1 (means going to the largest possible value, CDF will converge to 1).

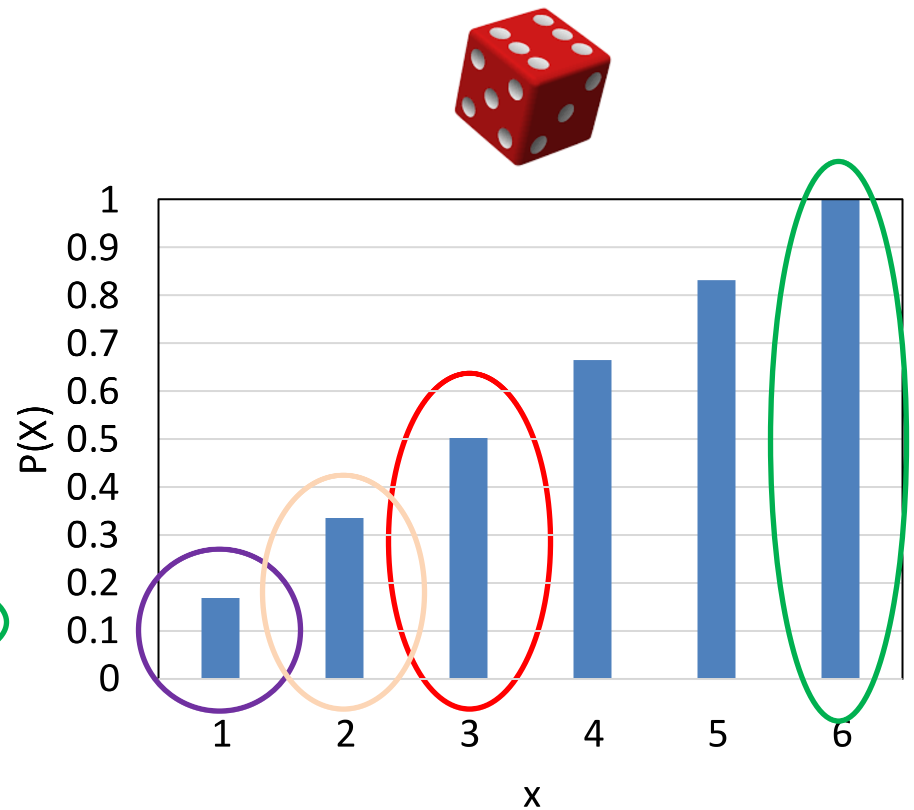
- Note: $$P_X(b) = \sum_{x \leq b} p_X(x)$$

$$P_X(a < x \leq b) = P_X(b) \text{-} P_X(a)$$

PMF vs CDF



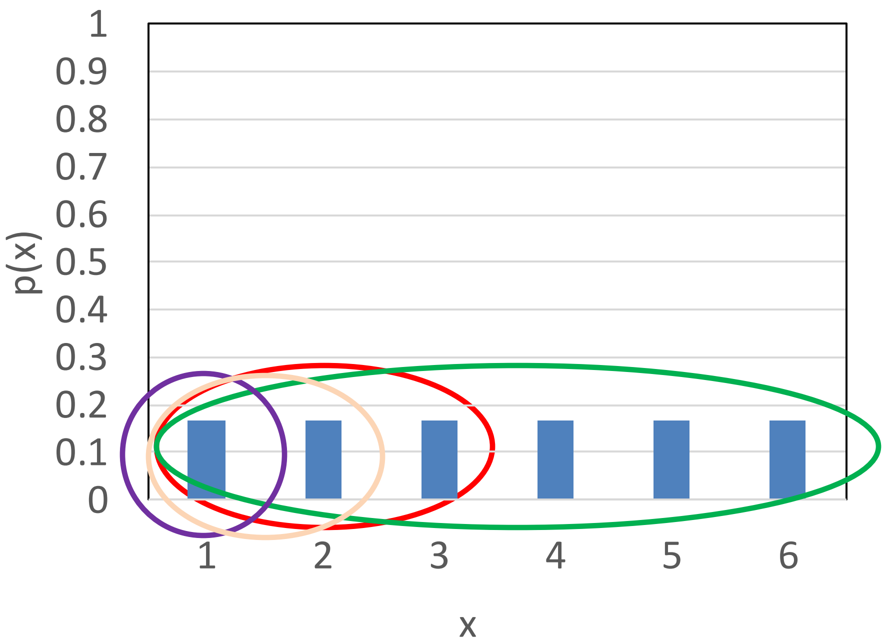
PMF

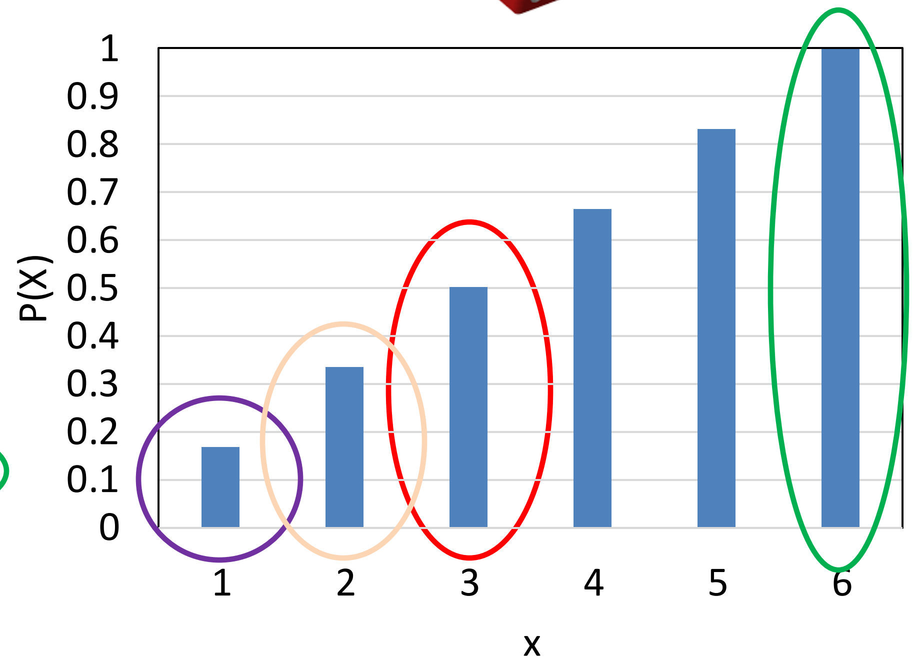


CDF

PMF vs CDF

To be clear, we are rolling a single dice in both examples. If you look at the CDF of 3, it is 0.5 which is the sum of rolling a 1, 2, or 3 = 1/6 + 1/6 + 1/6





PMF

CDF

Example 2

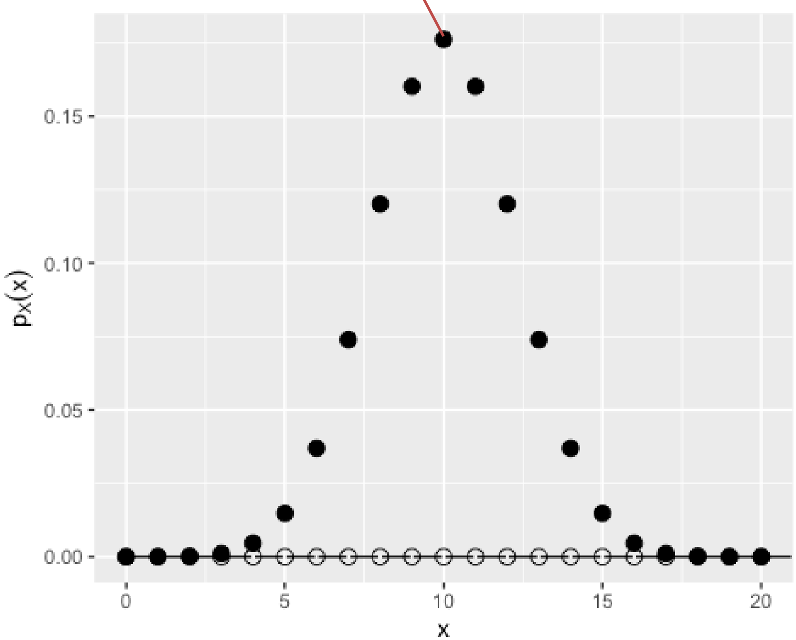
PMF vs CDF

X is number of "heads" in 20 flips of a fair coin

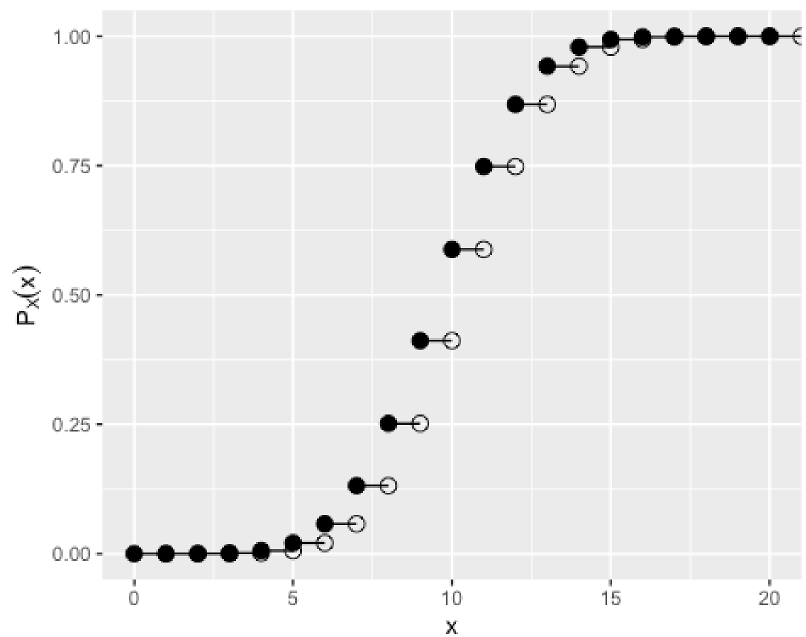
$x \in [0,20]$

Probability of k successes (heads) in n trials (flips): $\binom{n}{k} p^k (1-p)^{n-k} = \frac{n!(p)^k(1-p)^{n-k}}{(k)!(n-k)!}$

$$\boxed{P(X=10) = \binom{20}{10} (0.5)^{10}(0.5)^{10} = 184756 \cdot (0.5)^{20} \approx 0.1762}$$



PMF



CDF

3.4: Continuous Random Variables

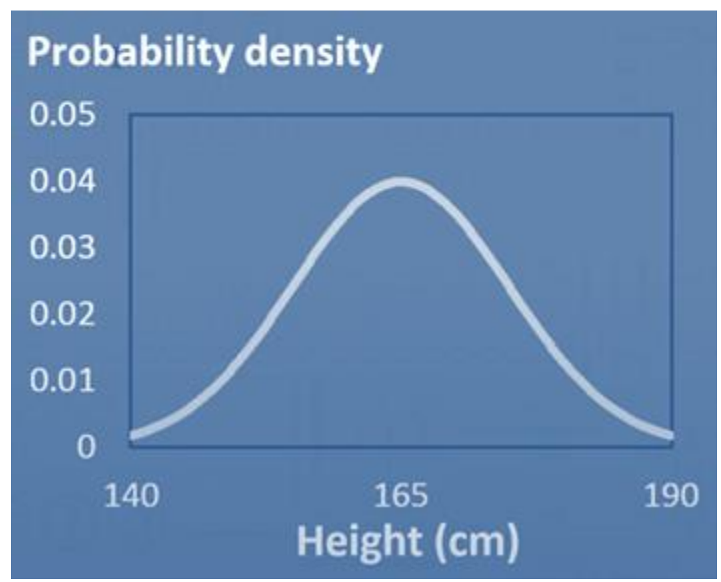
Continuous Random Variables

- Continuous random variables take values in the intervals of $\mathbb{R}{\geq}0$ (e.g. person's height in cm).
- The PMF does not exist: $\mathrm{P}(\mathrm{X} = \mathrm{x}) = 0$ for all x
- However, $\mathrm{P}\big(\mathrm{X} \in (\mathrm{a}, \mathrm{b})\big) \neq 0$.
- Probability Density Function (PDF) is defined as $f_X(x)$ where:
- Requirement:

$$\mathrm{P}(\mathrm{a} \leq \mathrm{X} \leq b) = \int_a^b f_X(x)\,dx \qquad , \qquad \forall x f_X(x) > 0 \;, \int_{-\infty}^{\infty} f_X(x) = 1$$

Example

PDF for person's height



Cumulative Distribution Function (CDF)

- For a continuous random variables X, $\mathrm{F}_X(\mathrm{x})$ gives

  $$\mathrm{P}(\mathrm{X} \leq x) = \mathrm{P}(X \in (-\infty, x])$$

- Requirement:

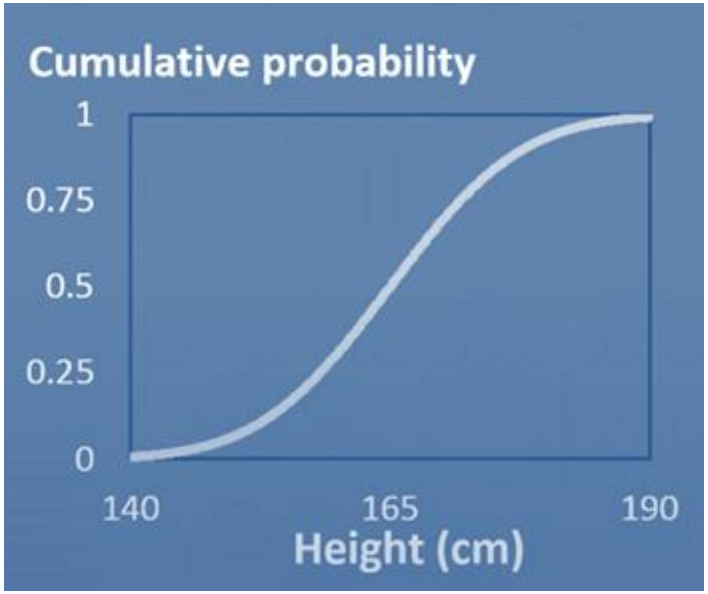
  $$F\ is\ nondecreasing \quad , \quad sup_x(F_X(x)) = 1$$

- Note :

  $$F_X(x) = \int_{-\infty}^{x} f_X(x)\,dx \quad , \quad P(x_1 < X \leq x_2) = F_X(x_2) \text{-} F_X(x_1)$$

Example

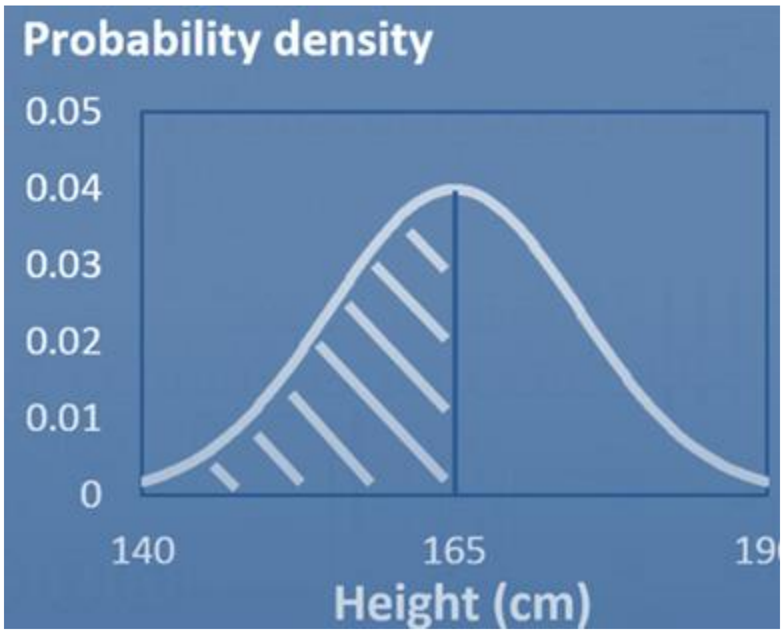
CDF for person's height



Example

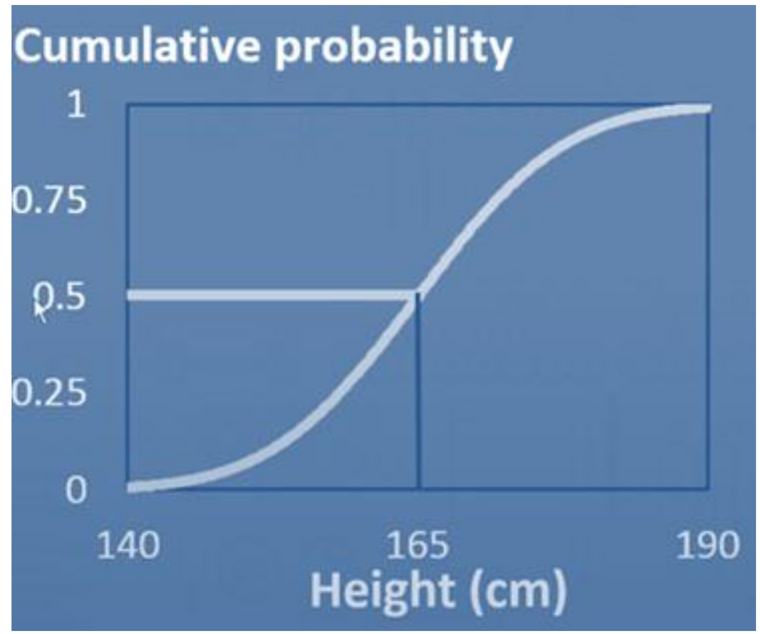
PDF vs CDF

$$F_X(x) = \int_{-\infty}^{x} f_X(x)\, dx$$



PDF

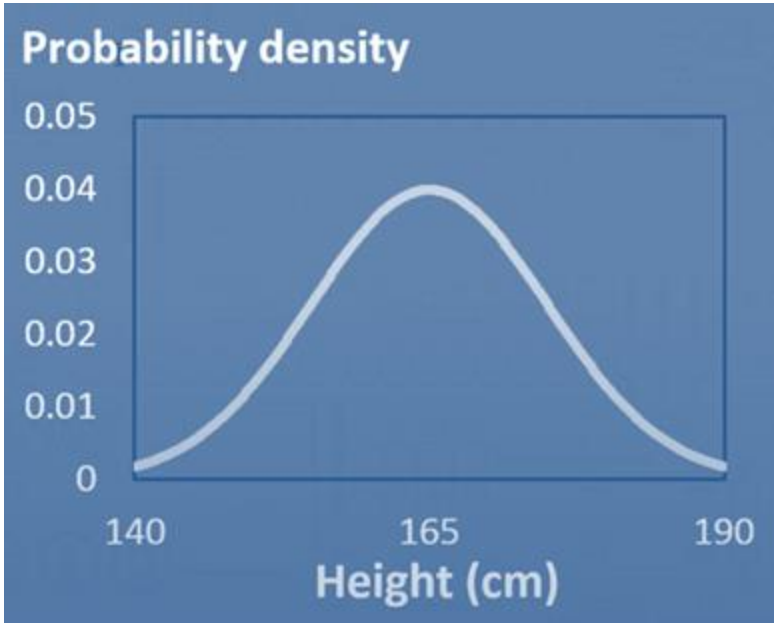
Area

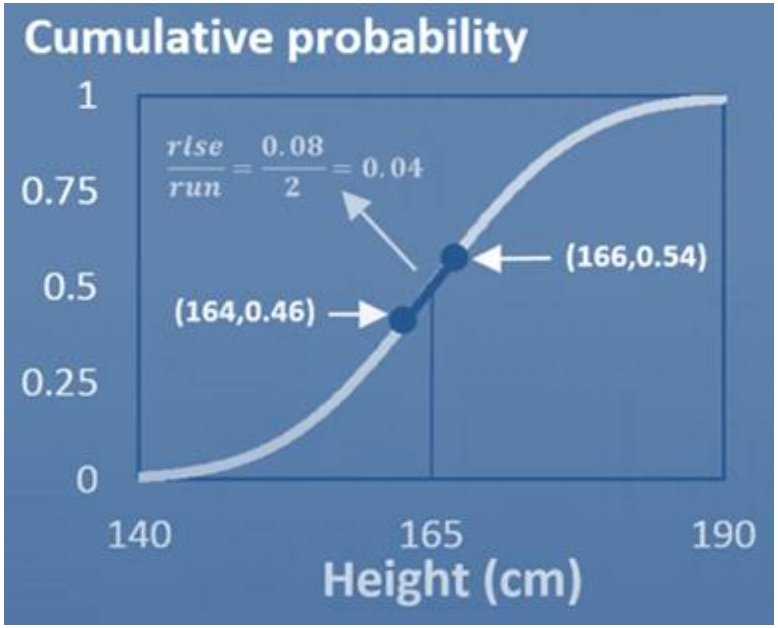


CDF

Example

PDF vs CDF





Gradient

PDF

$$f_X(x) = \frac{dF_X(x)}{dx}$$

CDF

Higher density ≡ Higher gradient

3.5: Random Vectors & Joint Distributions

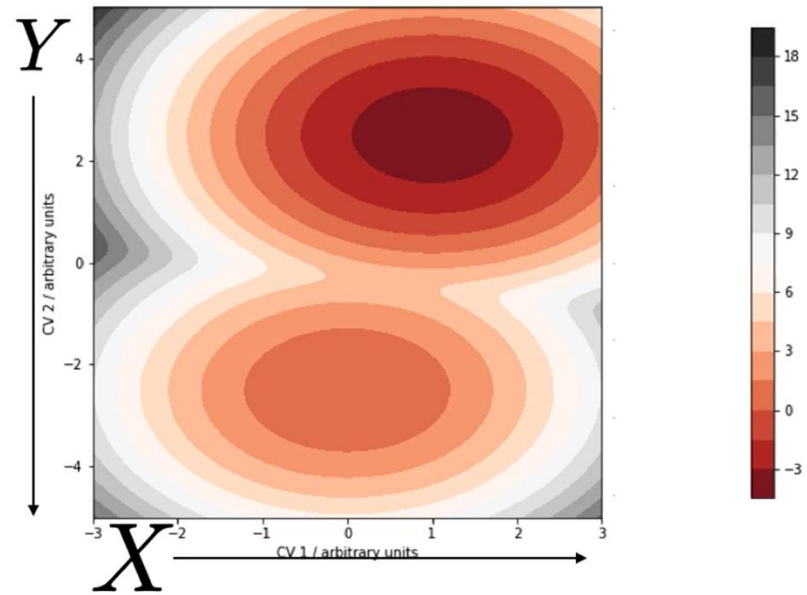
Joint Density

- Two random variables X and Y have a **joint distribution** if their realizations come together as a pair $(x_1, y_1), (x_2, y_2), \ldots$
- (X, Y) is a **random vector**.
- Joint Cumulative Distribution Function ($F_{X,Y}$):

$$P(X \leq b, Y \leq d) = F_{X,Y}(b, d)$$

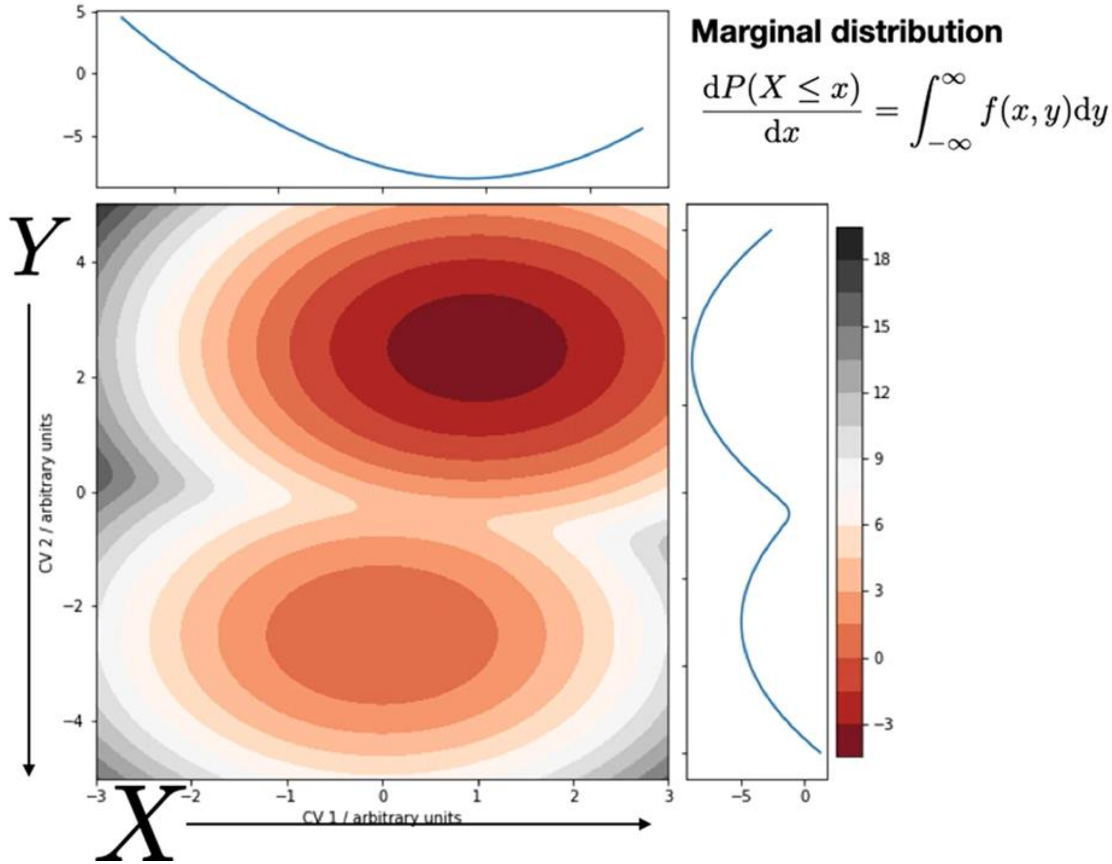
- Joint Probability Density Function ($f_{X,Y}(x, y)$):

$$P(X, Y \in \mathcal{A} \subseteq \chi, \gamma) = \iint_{\mathcal{A}} f_{X,Y}(x, y)\,dx\,dy$$



Marginal Density

- (X, Y) is a **Random Vector.**
- A Marginal distribution considers a subset of variables.
- If we ignore X, the distribution of Y is its marginal distribution. (Same for X but vice-versa.)

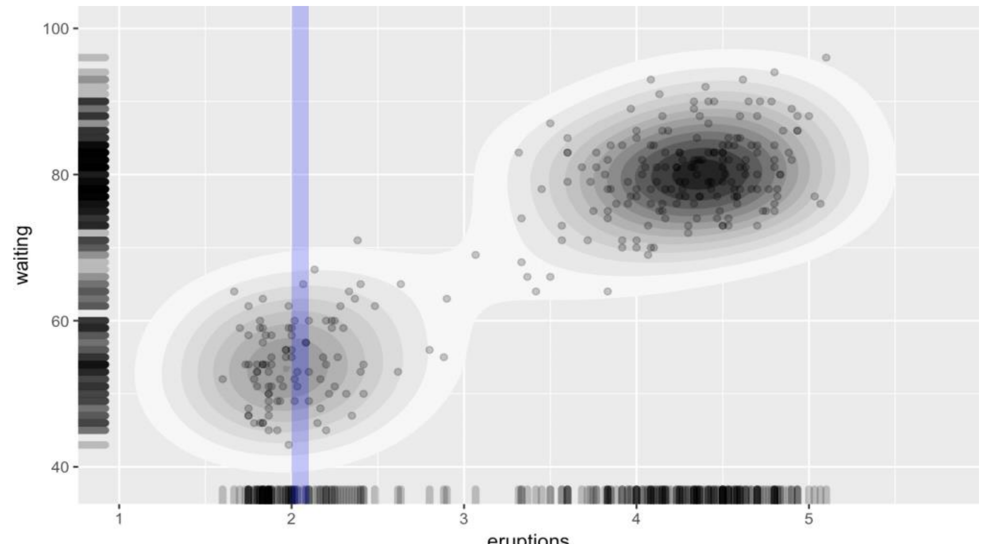


Conditional Distributions

- (X, Y) is a **Random Vector**.
- A conditional distribution fixes a subset of variables.
- Suppose we only look at Y values for which $X = x$
- This is a new random variable, which we write as $Y|X = x$
- The distribution describing this random variable is the conditional distribution of Y *given* $X = x$
- Divide joint distribution by marginal distribution

- The probability of event Y conditioned on knowing event X (probability of Y given X) is defined as:

Bayes' Rule: $P(Y|X) = \dfrac{P(Y, X)}{P(X)}$



Expectations

- The expected value of a discrete random variable X is denoted

$$E(X) = \sum_{x \in \chi} x.\, p_X(X = x)$$

- The expected value of a continuous random variable X is denoted

$$E(X) = \int_{x \in \chi} x.\, f_X(X = x)dx$$

- The expected value can be thought of as the "average" value attained by the random variable.
- $E(X)$ is called the mean of distribution, often denoted as μ or μ_X (measure of location of the distribution)

- We also define expected value for a function of a random variable. If g is a function (for example, $g(x) = x^2$), then the expected value of $g(X)$ is

$$E(g(X)) = \sum_{x \in \chi} g(x)P(X = x) = \sum_{x \in \chi} x^2 P(X = x)$$

Numerical Example

Netflix had a survey of 500 subscribers to determine favorite shows.

$$P(Male\ \cap GoT) = 0.24$$

Joint Probability

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

$Sums\ to\ 1$

Joint Probability Distribution

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

$P(GoT) = 0.4$

Marginal Probability

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

$Sums\ to\ 1$

Marginal Probability Distribution

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

$Sums\ to\ 1$

Marginal Probability Distribution

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

John just got the Netflix subscription, what is the chance that his favorite show will be GoT?

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)}$$

$$P(GoT|Male) = \frac{0.24}{0.54} = 0.4444$$

Conditional Probability

÷

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

	Female	Male	Total
Game of Thrones	0.16	0.24	0.4
West world	0.2	0.05	0.25
Other	0.1	0.25	0.35
Total	0.46	0.54	1

Probability Distribution

Numerical Example

Independency indicator:

Independent if $P(X|Y) = P(X)$

$P(Show|Male)$

	Female	Male	Total
Game of Thrones	80	120	200
West world	100	25	125
Other	50	125	175
Total	230	270	500

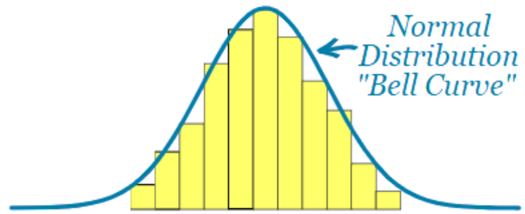
	Female	Male	Total	CP
Game of Thrones	0.16	0.24	0.4	0.44
West world	0.2	0.05	0.25	0.093
Other	0.1	0.25	0.35	0.463
Total	0.46	0.54	1	1

Probability Distribution

3.6: Distributions & Likelihood

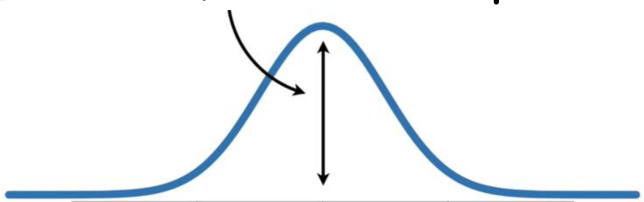
Normal (Gaussian) Distributions

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}}\, e^{-\frac{(X-\mu_X)^2}{2\sigma_X^2}}$$

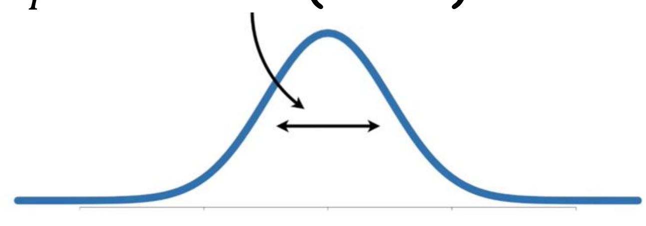


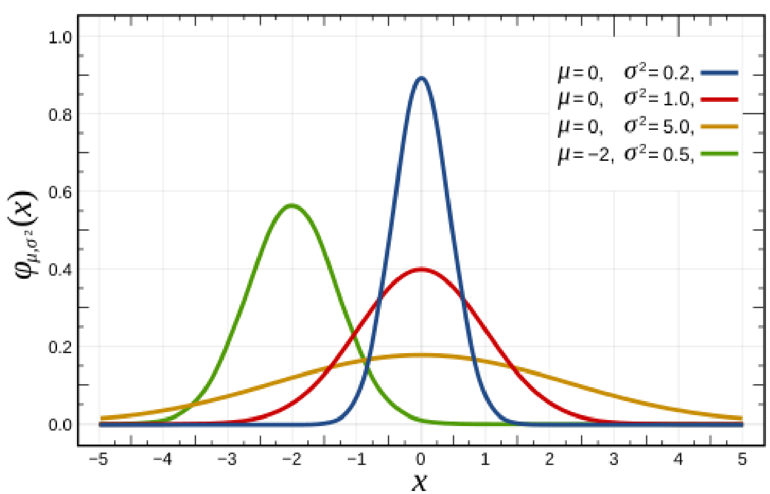
Normal distribution is defined by two parameters:

μ_Y: mean (location/expectation)



σ_Y^2: variance (scale)





Maximum Likelihood Estimation (MLE)

- MLE: defining a likelihood function for calculating the conditional probability of observing the data sample given a probability distribution and distribution parameters.

- This estimator provides the foundation for many ML algorithms, including linear regression and logistic regression (predicting numeric values and class labels respectively), as well as deep learning and artificial neural networks.
- The MLE is a method that finds the values for the parameters of a model that maximizes the likelihood for a given set of data.

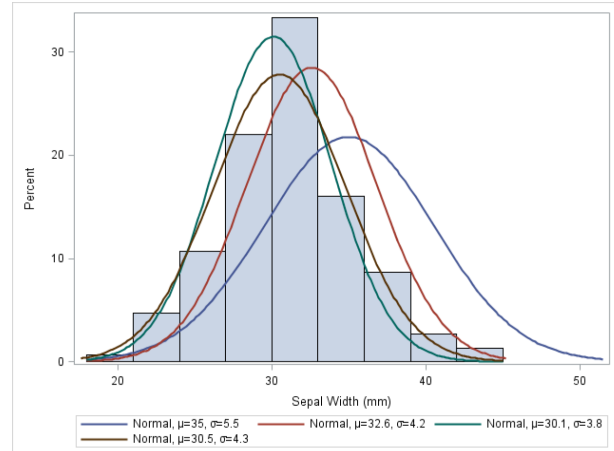
- For example, given a sample of observation (X) from a domain ($x_1, x_2, x_3, \ldots, x_n$)

- There are many techniques for solving this problem, although two common approaches are:
  - Maximum a Posteriori (MAP), a Bayesian method.
  - Maximum Likelihood Estimation (MLE), frequentist method

MLE is more common in machine learning because it is simpler, more efficient, and scales well with big data.

Maximum Likelihood Estimation (MLE)

- Seek a set of parameters that maximizes the likelihood of the data.

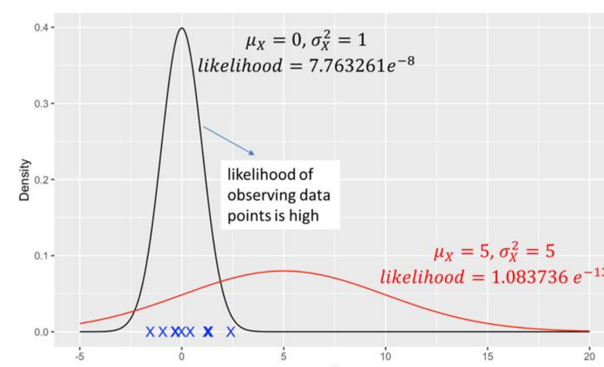


Likelihood:

- Given a distribution with parameters θ, how likely is my data?

- It is common to assume training examples are independent and identically distributed,(i.i.d.), then the probability of seeing all realizations is the product of probability of each realization.

$$\mathcal{L}(\theta; x_1, x_2, \dots, x_n) = \begin{cases} \prod_i f_X(\theta; x_i) \; if \; Continuous \\ \prod_i p_X(\theta; x_i) \;\; if \; Discrete \end{cases}$$



Log Likelihood

- Data Scientists mostly deal with log likelihoods.

- Likelihood is a product of many probabilities, each ≤ 1. For large n, the product becomes extremely small very fast (underflow). Taking log turns products into sums, which are numerically stable. Logs get negative but grow much more slowly.

  - Maximizing the likelihood $\equiv$ maximizing the log likelihood

$$a > b \quad \Leftrightarrow \quad \log(a) > \log(b)$$

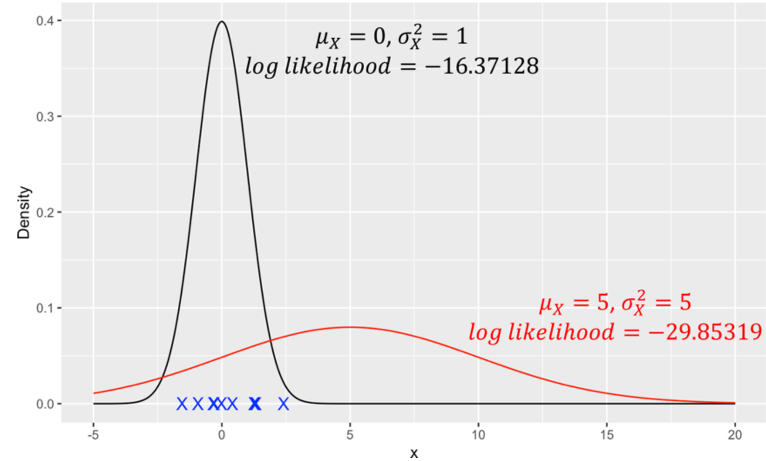
  - Logs turn **products** of probabilities into **sums** of log-probabilities.

$$\log(a \times b) = \log(a) + \log(b)$$

  - For exponential-family distributions, log likelihood has a simpler functional form.

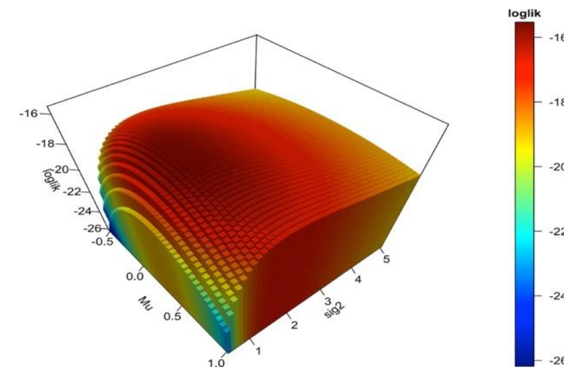
$$\mathcal{L}(\theta; x_1, x_2, \dots, x_n) = \begin{cases} \sum_i \log(f_X(\theta; x_i)) & \textit{if Continuous} \\ \sum_i \log(p_X(\theta; x_i)) & \textit{if Discrete} \end{cases}$$

Log Likelihood



$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$$

$$\log\left(f_X(x)\right) = \log\left(\frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}\right)$$



$$\log\left(f_X(x)\right) = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\sigma_X^2) - \frac{1}{2}\frac{(x-\mu_X)^2}{\sigma_X^2}$$

$$\mathcal{L}(\mu_X, \sigma_X^2; x_1, x_2, \ldots, x_n) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma_X^2) - \frac{1}{2}\frac{\sum_{i=1}^{n}(x_i - \mu_X)^2}{\sigma_X^2}$$

Maximum Likelihood Estimation (MLE)

$$\mathcal{L}(\mu_X, \sigma_X^2; x_1, x_2, \dots, x_n) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma_X^2) - \frac{1}{2}\frac{\sum_{i=1}^{n}(x_i - \mu_X)^2}{\sigma_X^2}$$

What μ_X gives the highest likelihood?

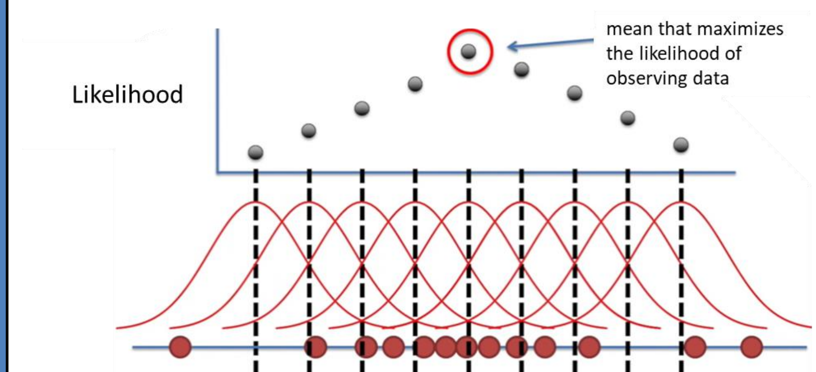
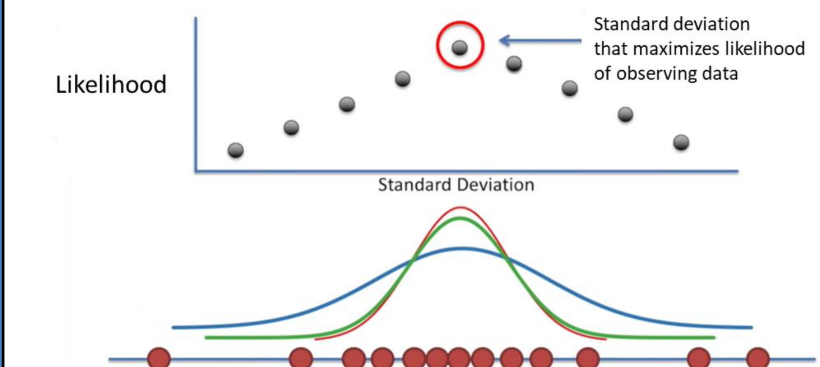
$$\frac{\partial \mathcal{L}}{\partial \mu_X} = \frac{1}{\sigma_X^2}\sum_{i=1}^{n}(x_i - \mu_X)^2$$

$$\frac{\partial \mathcal{L}}{\partial \mu_X} = 0 \quad \Leftrightarrow \mu_X = \frac{\sum_{i=1}^{n} x_i}{n}$$

What σ_X^2 gives the highest likelihood?

$$\frac{\partial \mathcal{L}}{\partial \sigma_X^2} = -\frac{n}{2\sigma_X^2} + \frac{\sum_{i=1}^{n}(x_i - \mu_X)^2}{2\sigma_X^4}$$

$$\frac{\partial \mathcal{L}}{\partial \sigma_X^2} = 0 \quad \Leftrightarrow \sigma_X^2 = \frac{\sum_{i=1}^{n}(x_i - \mu_X)^2}{n}$$

Mean	
Standard deviation	

3.7: Maximum Likelihood Estimation (MLE) Method for Linear Regression

Linear Regression

- Regression methods learn a model: $\hat{y} = f(x) = \theta_0 + \theta_1 x$
- Recall: How are θ_0 and θ_1 determined?
- We are estimating Conditional Expectation $E(\hat{y}|X)$

$$MLE(\theta) = \max_{\theta}(\mathcal{L}(\hat{y}|X))$$

- Model y is normally distributed with mean of $\theta_0 + \theta_1 x$ and standard deviation of σ^2:

$$y = f(x) + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2) \quad \rightarrow \quad y_i \sim N(\underbrace{f(x_i)}_{\text{mean}}, \underbrace{\sigma^2}_{\text{variance}})$$

- Regression's Log Likelihood:

$$\mathcal{L}(\theta_0, \theta_1, \sigma_\varepsilon^2; (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$$
$$= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma_\varepsilon^2) - \frac{1}{2}\frac{\sum_{i=1}^{n}(y_i - (\theta_0 + \theta_1 x_i))^2}{\sigma_\varepsilon^2}$$

MLE in Linear Regression

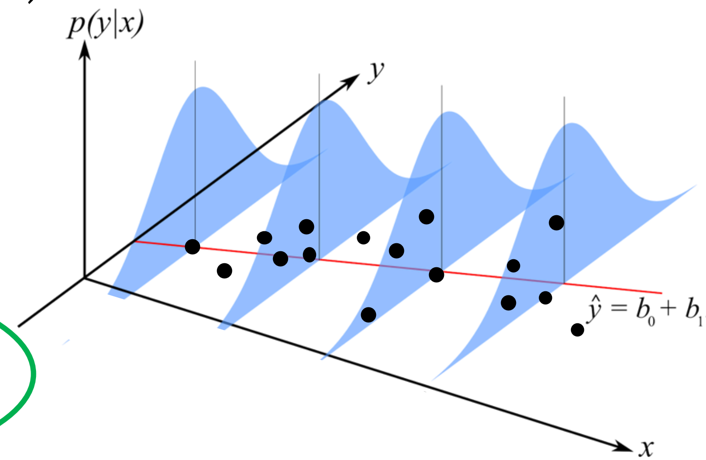
- Maximizing the Log Likelihood:

$$\max_{\theta}\left(-\frac{n}{2}\log(2\pi)-\frac{n}{2}\log(\sigma_{\varepsilon}^{2})-\frac{1}{2}\frac{\sum_{i=1}^{n}(y_i-(\theta_0+\theta_1 x_i))^2}{\sigma_{\varepsilon}^{2}}\right)$$

$$\equiv \max_{\theta}\sum_{i=1}^{n}-(y_i-(\theta_0+\theta_1 x_i))^2$$

$$\equiv \min_{\theta}\sum_{i=1}^{n}(y_i-(\theta_0+\theta_1 x_i))^2 \equiv \min_{\theta}\sum_{i=1}^{n}\big(y_i-f(x)\big)^2$$

Least Squares



- MLE is equivalent to minimizing least-squared error!
- This is a probabilistic interpretation or Bayesian explanation of the least-squared error solution and why did we choose squared error for defining a loss function.